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UE23CS342BA1
SUPPLY CHAIN MANAGEMENT (SCM)
For Engineers

PROJECT REPORT
on
<Problem No. _1__>

< Order & Inventory Visibility Dashboard using Supply Chain Data >

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INTRODUCTION

Problem No. & Description

This project addresses **Problem P1 – Order & Inventory Visibility**.

Companies face challenges in:

- tracking customer orders and shipment status
- identifying delays in deliveries
- monitoring inventory levels in real time

Due to lack of visibility:

- deliveries are delayed
- stockouts occur
- decision-making becomes inefficient

This project provides a **data-driven dashboard** to improve visibility across orders, shipments, and inventory.

SCOPE & OBJECTIVES

Scope

- Track customer orders and shipment progress
- Monitor inventory levels
- Identify delayed deliveries and low stock items

Objectives

- Build an interactive dashboard using Power BI
- Provide insights using KPIs and charts
- Improve supply chain decision-making
- Apply machine learning for demand prediction

SOLUTION WITH KEY FEATURES

The solution is an **interactive dashboard** integrating order, shipment, and inventory data.

Key Features

- Real-time tracking of orders and shipment status
- Identification of delayed orders
- Monitoring of inventory levels and low stock items
- Visualization of order trends over time

KPIs TRACKED

- **Total Orders** – total number of orders placed
- **Delayed Orders** – number of orders delivered late
- **Low Stock Items** – products below reorder level

DATA INPUTS

a) Source

- Self generated supply chain dataset

b) Description

The dataset is divided into three tables:

Orders

- order_id
- product_id
- order_date
- quantity
- status

Inventory

- product_id
- product_name
- stock_level
- reorder_level

Shipments

- order_id
- shipment_status
- delay_days

c) Database Design

The data is structured into three related tables:

- Orders
- Inventory
- Shipments

Each table represents a different aspect of the supply chain.

d) Relationships – Schema

- Orders connected to Inventory using **product_id**
- Orders connected to Shipments using **order_id**

This allows integration of all data for analysis.

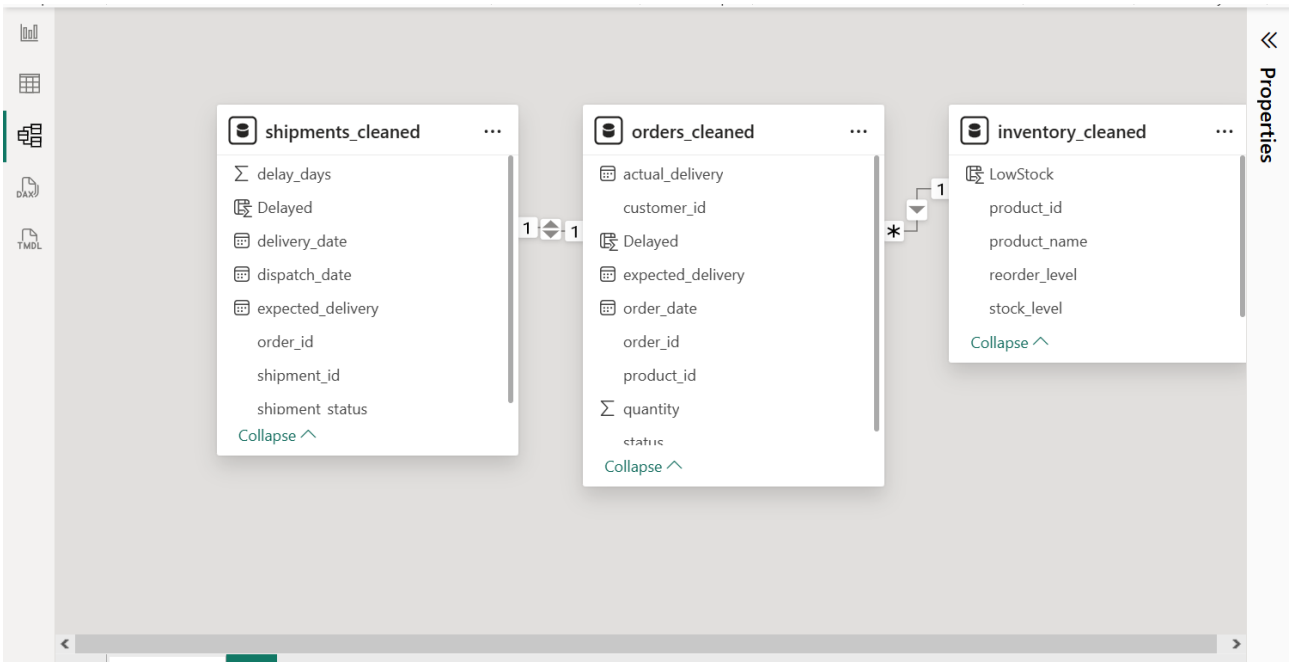

fig 1.0 Relationship model from PowerBI

e) DBMS

No traditional DBMS is used.
Data is handled using:

- CSV files
- Power BI data model

DATA OUTPUTS / VISUALIZATION

a) Dashboard Design & Graphs

The dashboard includes:

- **Card visuals (KPIs)**
  - Total Orders
  - Delayed Orders
  - Low Stock Items
- **Line Chart**
  - Orders Over Time
- **Pie Chart**
  - Shipment Status (On-time vs Delayed)
- **Bar Chart**
  - Inventory Levels by Product

b) Interactivity

The dashboard is interactive and includes:

- Date filters
- Product filters
- dynamic updates of visuals

c) Report Samples

The dashboard provides visual insights such as:

- trends in order volume
- distribution of delayed shipments
- comparison of stock levels across products

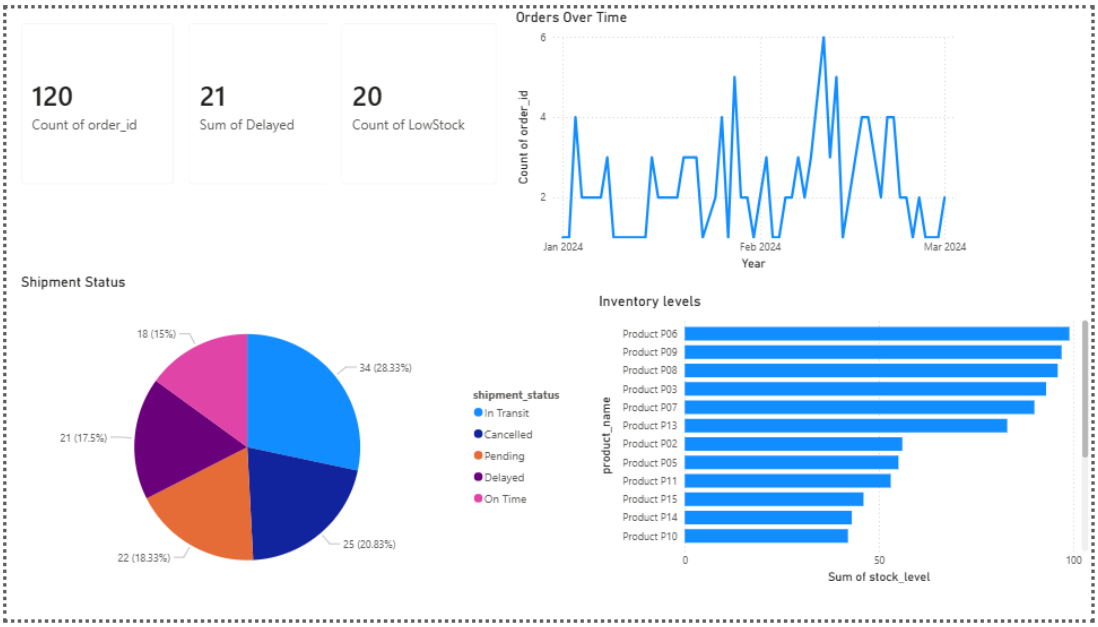

fig 1.2 PowerBI dashboard

MACHINE LEARNING

a) ML Algorithm Used

Linear Regression

Purpose

- To predict future demand based on past order data

Application

- Uses order history (date and quantity)
- Predicts demand for upcoming days

b) How ML Enhances the Solution

- improves demand forecasting
- helps in inventory planning
- reduces stockouts and overstocking
- supports data-driven decision-making

c) ML Tools Used

- Python
- Libraries:
    - pandas
    - scikit-learn
    - matplotlib

d) **Installation**

pip install pandas scikit-learn matplotlib

RESULTS, CONCLUSION AND FUTURE WORK

Summary Discussion

The dashboard successfully integrates order, shipment, and inventory data into a single system. It provides clear visibility into supply chain operations.

Business Insights and Impacts

- delayed shipments are easily identified
- low stock products can be monitored
- order trends help in forecasting demand
- improves efficiency of supply chain decisions

Conclusion

The project demonstrates how data visualization and ML can improve supply chain visibility. It helps organizations monitor operations and take informed decisions.

Future Work

- integration with real-time data sources
- advanced ML models for better forecasting
- automated alerts for low stock and delays

REFERENCES

- Supply Chain Dashboard Resources
- Power BI Documentation
- scikit-learn Documentation
- Course materials provided